Chinook Digital Media — Customer & Revenue Analytics

Executive Summary

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OVERVIEW

An end-to-end analytics portfolio built on the Chinook digital media database: three SQL case studies, a multi-part Python investigation, and three interactive Tableau dashboards. Each phase builds on the last. The through-line is a practical one: where should a marketing team focus its budget, and why?

KEY FINDINGS

1. Three markets carry the business — and that's a risk.

The USA, Canada, and France account for 44% of total revenue across 24 countries. The bottom 15 markets each depend on a single customer. Losing one means that market goes to zero. The recommendation: grow mid-tier markets (Germany, UK, Brazil) to reduce geographic concentration.

2. Customer tiers are compressed because the product pricing is too narrow.

72% of revenue sits in a single value tier (Silver). The cause is structural: with tracks priced at $0.99 and $1.99, there isn't enough price variance to create meaningful spend separation. Segmentation works, but upsell opportunities require product or pricing changes, not just marketing campaigns.

3. Four genres drive 73.5% of revenue — trend direction decides if that's a problem.

Rock, Latin, Metal, and Alternative & Punk dominate the catalog. If those four are growing, concentration is fine. If they're flat or declining, there's no strong second tier to absorb the loss. Sci Fi & Fantasy leads in catalog efficiency ($1.53/track vs. $0.64 for Rock), pointing to unmet demand in a small but high-performing genre.

4. Customers buy on a ~32-day cycle, and the intervention window is narrow.

The median repurchase interval is 32 days with high natural variance. 57.6% of repeat purchases happen within 6 weeks. A proactive engagement campaign at 14–21 days and a win-back trigger at 40–100 days would cover the two highest-leverage windows in the customer lifecycle.

5. Retention follows the same curve regardless of when a customer joined.

Every quarterly cohort decays at the same rate (±7 percentage points across 26 cohorts over 6 years). The steepest drop is at Q+4 (–14.5pp). Half of all churn happens by Q+5. The YoY growth deceleration isn't a retention problem — it's an acquisition or product ceiling. Retention investment should target the first 12 months, where the highest share of churn is still preventable.

TOOLS & APPROACH

SQL: Revenue segmentation, CLV tiering, catalog benchmarking (CTEs, window functions, multi-table joins) **Python:** Purchase timing analysis, cohort retention, time series (pandas, matplotlib, seaborn) **Tableau:** Three interactive dashboards with cross-filtering, published to Tableau Public

AI (Claude) was used as a collaborative tool throughout. The full workflow — structured prompts, critical evaluation, iterative refinement, and every judgment call — is documented in the repository.

View Interactive Dashboards on Tableau Public